

APPENDIX

Table I summarizes the complete Taguchi L9-based experimental conditions, including the four actuation configurations, factor levels, and repeated trials used for the energy analysis.

TABLE I

EXPERIMENTAL CONDITIONS BASED ON THE TAGUCHI L9 ORTHOGONAL ARRAY				
Run	Configs.	f(Hz)	T_{pre}(N)	P(psi)
1	NO PAM	0.75	30	0
2	NO PAM	0.75	30	0
3	NO PAM	0.75	30	0
4	NO PAM	0.75	45	0
5	NO PAM	0.75	45	0
6	NO PAM	0.75	45	0
7	NO PAM	0.75	60	0
8	NO PAM	0.75	60	0
9	NO PAM	0.75	60	0
10	NO PAM	1	30	0
11	NO PAM	1	30	0
12	NO PAM	1	30	0
13	NO PAM	1	45	0
14	NO PAM	1	45	0
15	NO PAM	1	45	0
16	NO PAM	1	60	0
17	NO PAM	1	60	0
18	NO PAM	1	60	0
19	NO PAM	1.25	30	0
20	NO PAM	1.25	30	0
21	NO PAM	1.25	30	0
22	NO PAM	1.25	45	0
23	NO PAM	1.25	45	0
24	NO PAM	1.25	45	0
25	NO PAM	1.25	60	0
26	NO PAM	1.25	60	0

27	NO PAM	1.25	60	0
28	PEA	0.75	30	5
29	PEA	0.75	30	5
30	PEA	0.75	30	5
31	PEA	0.75	45	13
32	PEA	0.75	45	13
33	PEA	0.75	45	13
34	PEA	0.75	60	20
35	PEA	0.75	60	20
36	PEA	0.75	60	20
37	PEA	1	30	13
38	PEA	1	30	13
39	PEA	1	30	13
40	PEA	1	45	20
41	PEA	1	45	20
42	PEA	1	45	20
43	PEA	1	60	5
44	PEA	1	60	5
45	PEA	1	60	5
46	PEA	1.25	30	20
47	PEA	1.25	30	20
48	PEA	1.25	30	20
49	PEA	1.25	45	5
50	PEA	1.25	45	5
51	PEA	1.25	45	5
52	PEA	1.25	60	13
53	PEA	1.25	60	13
54	PEA	1.25	60	13
55	SEA	0.75	30	5
56	SEA	0.75	30	5
57	SEA	0.75	30	5
58	SEA	0.75	45	13
59	SEA	0.75	45	13
60	SEA	0.75	45	13
61	SEA	0.75	60	20
62	SEA	0.75	60	20
63	SEA	0.75	60	20
64	SEA	1	30	13

65	SEA	1	30	13
66	SEA	1	30	13
67	SEA	1	45	20
68	SEA	1	45	20
69	SEA	1	45	20
70	SEA	1	60	5
71	SEA	1	60	5
72	SEA	1	60	5
73	SEA	1.25	30	20
74	SEA	1.25	30	20
75	SEA	1.25	30	20
76	SEA	1.25	45	5
77	SEA	1.25	45	5
78	SEA	1.25	45	5
79	SEA	1.25	60	13
80	SEA	1.25	60	13
81	SEA	1.25	60	13
82	SPEA	0.75	30	5
83	SPEA	0.75	30	5
84	SPEA	0.75	30	5
85	SPEA	0.75	45	13
86	SPEA	0.75	45	13
87	SPEA	0.75	45	13
88	SPEA	0.75	60	20
89	SPEA	0.75	60	20
90	SPEA	0.75	60	20
91	SPEA	1	30	13
92	SPEA	1	30	13
93	SPEA	1	30	13
94	SPEA	1	45	20
95	SPEA	1	45	20
96	SPEA	1	45	20
97	SPEA	1	60	5
98	SPEA	1	60	5
99	SPEA	1	60	5
100	SPEA	1.25	30	20
101	SPEA	1.25	30	20
102	SPEA	1.25	30	20

103	SPEA	1.25	45	5
104	SPEA	1.25	45	5
105	SPEA	1.25	45	5
106	SPEA	1.25	60	13
107	SPEA	1.25	60	13
108	SPEA	1.25	60	13